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Impact tool with front lubrication assembly

Abstract

An impact tool having a front lubrication assembly. The front lubrication assembly includes a lubrication port, a lubrication passage and at least one lubrication channel that directs a lubricant injected from a front end of the impact tool directly into an impact assembly of the impact tool. The lubrication passage extends through an anvil assembly of the impact tool, along an axis of rotation of the impact drive mechanism. The at least one lubrication channel extends away from the lubrication passage and delivers the lubricant to at the impact mechanism. The front lubrication assembly is accessed on a front end of the impact tool. The anvil assembly may be a split anvil assembly having an external anvil portion and an external anvil portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application claims priority under 35 U.S.C. § 119 (e) of U.S. Provisional Application Ser. No. 63/404,063, filed Sep. 6, 2022, and titled "Impact Tool with Split Anvil and Lubrication Port". U.S. Provisional Application Ser. No. 63/404,063 and the co-pending Non-Provisional application Ser. No. 18/239,362 titled "Impact Tool with Split Anvil," dated Aug. 29, 2023, are incorporated by reference herein in their entireties.

BACKGROUND

(1) Power tools are power tools configured to deliver a high torque output by storing energy in a rotating mass and delivering it suddenly through an output shaft to a fastener. In order to function properly, power tools may be regularly scheduled for maintenance.

Description

DRAWINGS

(1) The Detailed Description is described with reference to the accompanying figures. The use of the same reference numbers in different instances in the description and the figures may indicate similar or identical items.

(2) FIG. 1 is a partial cross-sectional view of a power tool having a dual-hammer impact mechanism and front lubrication port in accordance with example embodiments of the present disclosure.

(3) FIG. 2 is a partial cross-sectional side view of the anvil shown in FIG. 1, wherein the anvil is a split anvil in accordance with example embodiments of the present disclosure.

(4) FIG. 3 is a perspective cross-sectional view of an anvil front lubrication port in accordance with example embodiments of the present disclosure.

(5) FIG. 4 is a partial cross-sectional view of a power tool having a ball-and-cam impact mechanism and a front lubrication port in accordance with example embodiments of the present disclosure.

(6) FIG. 5 is a side cross-sectional view of the ball-and-cam impact mechanism from FIG. 4 cut along line 5-5 having an anvil front lubrication port in accordance with example embodiments of the present disclosure.

(7) FIG. 6 is a top cross-sectional view of the power tool from FIG. 4 cut along line 6-6 in accordance with example embodiments of the present disclosure.

DETAILED DESCRIPTION

(8) Although the subject matter has been described in language specific to structural features and/or process operations, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing the claims.

OVERVIEW

(9) Impact tools (e.g., impact wrenches, etc.) are designed to deliver a high torque output with minimal exertion by the user. Impact mechanisms include a rotating mass (e.g., a hammer) that stores energy in impact jaws. The impact jaws abruptly deliver the stored energy to an anvil connected to an output shaft, subjecting the anvil to repeated and sudden shock loading.

(10) In order for an impact tool to operate optimally, the impact mechanism is lubricated. Lubricants may reduce the heat generated by the impact of the impact mechanism jaws to the anvil, or the wear of an impact bearing to a cam shaft and hammer of the impact mechanism by creating a film between the two impacting surfaces. The lubricant reduces friction and improves the efficiency and performance of the impact tool. This lubrication is typically, but not always, a specially formulated grease that is installed at the factory. Overtime this lubrication tends to break down or migrate away from the areas of impact and wear, resulting in a dry impact assembly condition. This dry impact assembly condition may lead to premature wearing of mechanism parts resulting in a performance reduction or in more severe cases a stalling, or a locking up, condition rendering the tool inoperable.

(11) Impact tools may contain a grease fitting located either on the hammer case or at the rear of the tool for lubricating the impact mechanism. The downside of locating the grease fittings around the hammer case is that the injected grease lubricates the outside of the mechanism and not directly

on the impact jaws where it is needed. Tools with rear grease ports tend to do a better job of getting the lubrication where at the impact jaws but require a rotor of the drive mechanism to have a through hole, which is not always feasible. For impact tools that do not contain a grease fitting, or the location in need of lubrication is too internal for the grease to migrate efficiently to this area, the tool must be disassembled to manually apply grease directly to the areas of stress and wear in the impact mechanism.

(12) The impact tool described herein includes a front lubrication assembly that directs a lubricant injected from a front end of the impact tool directly into the main areas of stress in the impact mechanism (e.g., anvil jaws, hammer jaws, impact bearing, helical grooves, etc.) of the impact mechanism assembly. The front lubrication assembly includes a lubrication passage that extends through the anvil assembly and along an axis of rotation of the impact tool. The lubrication passage splits the flow of the lubricant into lubrication channels that extend away from the lubrication passage and deliver the lubricant flow to the areas of stress and wear in the impact mechanism.

(13) The anvil assembly may be a split anvil assembly having an internal anvil portion fixed inside a housing of the impact tool and an external anvil portion outside the housing. The external anvil portion is removably connected to the internal anvil portion. The front lubrication assembly may be accessed when the external anvil portion is disengaged from the internal anvil portion and an inlet of the lubrication passage is accessible through an internal anvil portion cavity.

(14) The external anvil portion may be selected from a plurality of replaceable anvil attachments, including but not limited to anvils with different drive sizes, socket extensions, custom sockets, etc. that are interchangeable without disassembling the impact tool.

Detailed Description of Example Embodiments

(15) Referring generally to FIGS. **1** through **3**, an impact tool having a front lubrication assembly **140** is described. FIG. **1** shows an illustrative embodiment of an impact tool assembly **100** in accordance with the present disclosure. The impact tool includes a housing **102** having a front end **101** and a rear end **103**. The impact tool assembly **100** includes a hammercase **104** that houses an impact assembly **110**. The housing **102** includes a drive mechanism **105** that rotates a hammer **106** of the impact assembly **110** around an output axis **100A**. The output axis **100A** extends from the front end **101** to the rear end **103**. The housing may include a gear set assembly **107** connecting the drive assembly **105** with the hammer **106**.

(16) In embodiments, the drive mechanism **105** comprises a pneumatic (compressed air) motor powered by a source of compressed air (not shown). However, it is contemplated that the impact tool assembly **100** may also include an electric motor (not shown) powered by a power source such as a removable battery, an internal battery, or an external power source via an electric cord. In other embodiments, the impact tool assembly **100** may be hydraulically operated.

(17) The hammer **106** includes at least one hammer jaw **112** extending radially from the axis **100A**. The impact assembly **110** further includes an anvil assembly **115**, for example, the one shown in FIGS. **2** and **3**. The anvil assembly **115** includes at least one anvil jaw **132** configured to be repeatedly struck by the at least one hammer jaw **112**. As the hammer **106** continuously and intermittently impacts against the at least one anvil jaw **132** anvil assembly **115** continuously rotates. An output shaft **125** extends from the anvil assembly **115** and may receive a connector, a socket, or other device that engages a fastener (e.g., a bolt, a nut, a screw, etc.) to be tightened or loosened as the anvil assembly **115** rotates with respect to the output axis **100A**.

(18) In example embodiments, the anvil assembly **115** may be a split anvil assembly. The split anvil assembly may include an external anvil portion **120** and an internal anvil portion **130**, where the internal anvil portion **130** is fixed inside the hammercase **104** and the external anvil portion extends longitudinally from the front end **101** and is removably attached to the internal anvil portion **130**. In this embodiment, the external anvil portion **120** extends longitudinally from the front end **101** outside of the hammercase **104** and the housing **102**. The internal anvil portion **130** includes at least one anvil jaw **132** configured to be repeatedly struck by the at least one hammer jaw **112**. As the

hammer **106** continuously and intermittently impacts against the internal anvil portion **130** of the split anvil assembly **115**, the external anvil portion **120** continuously rotates when the external anvil portion **120** is engaged and secured to the internal anvil portion **130**. An output shaft **125** extends from the external anvil portion **120** and may receive a connector, a socket, or other device that engages a fastener (e.g., a bolt, a nut, a screw, etc.) to be tightened or loosened.

(19) The impact tool assembly **100** includes a front lubrication assembly **140**. The front lubrication assembly **140** includes a lubrication passage **141** defined through the anvil assembly **115** and extending axially along axis **100A**. The front lubrication assembly **140** may include a grease fitting **145** having a ball **146** and a spring **148**. In example embodiments, the ball **146** is pushed against the spring **148** by an outside pressure (e.g., a grease gun) and a lubricant is injected into the impact assembly **110**. The lubricant injected by a user (e.g., grease) passes into a channel **147** of the grease fitting **145** and flows into the lubrication passage **141** and through at least one lubrication channel **142**, and directly to the impact jaws (e.g., hammer jaw, anvil jaw) of the impact assembly **110**. The at least one lubrication channel **142** extends away from the lubrication passage **141**. For example, the at least one lubrication channel **142** may extend radially away from, or perpendicular to, the lubrication passage **141**. In other embodiments (not shown) the at least one lubrication channel **142** may extend at an angle between zero degrees (0°) and ninety degrees (90°) with respect to the lubrication passage **141** or the axis **100A**. The grease fitting **145** may be fixedly attached to the anvil assembly **115** by a tapered thread at the inlet of the lubrication passage **141**, as a straight push-fit arrangement, or by another arrangement.

(20) FIG. 1 shows an example embodiment of the anvil assembly **115**, having the external anvil portion **120** and the internal anvil portion **130**, connected to the impact tool **100**. The hammercase **104** includes a bushing **114** and a cover ring **116** holding the internal anvil portion **130** in place. The bushing **114**, the cover ring **116**, and the internal anvil portion **130**, respectively include an access port **131**. The internal anvil portion **130** defines an internal anvil portion cavity **135**. The internal anvil portion cavity **135** includes an internal anvil cavity wall **130a**. The internal anvil cavity wall **130a** further defines the lubrication passage **141** of the front lubrication assembly **140** and at least one lubrication channel **142**. A lubrication port inlet **144** is disposed within the internal anvil portion cavity **135** at an opening of the lubrication passage **141**.

(21) In example embodiments, the external anvil portion **120** defines an external anvil portion cavity **126** including a retaining cavity **128**, and a retaining orifice **121**. The external anvil portion cavity **126** houses a retaining pin **124**. The retaining pin **124** is configured to engage with the access port **131** of the internal anvil portion **130**, thereby effectively locking the external anvil portion **120** and the internal anvil portion **130**. Upon retraction of the retaining pin **124**, the external anvil portion **120** disengages with the internal anvil portion **130**, exposing the internal anvil portion cavity **135**.

(22) The retaining cavity **128** houses a biasing member **122** that retains the retaining pin **124** within the retaining orifice **121**. In embodiments, when the external anvil portion **120** is engaged with the internal anvil portion **130**, the biasing member **122** biases the retaining pin **124** outward towards the access port **131** of the internal anvil portion **130**, locking the two portions of the split anvil assembly **115** together. In order to separate the external anvil portion **120** and the internal anvil portion **130**, the retaining pin **124** may be depressed with an elongated tool (not shown) until the retaining pin **124** is fully depressed out of the access port **131**. The output shaft **125** of the split anvil assembly **115** can be replaced by inserting an appropriately sized elongated tool (e.g., a screwdriver) through the access port **131** and depressing the retaining pin **124**.

(23) It should be understood that other attachment methods may be used to retain the external anvil portion **120** into the internal anvil portion cavity **135**. Other retaining assemblies may include, but are not limited to, actuation buttons to actuate the retaining pin **124**, retaining caps, retaining rings, retractable ball detent mechanisms on at least one of the internal anvil portion and/or the external anvil portion, hog rings, among others.

(24) In the embodiment shown in FIG. 2, the external anvil portion **120** includes external splines **123** defined around the circumference of the outer surface of the external anvil portion **120**. The internal anvil portion **130** may also include internal splines **133** defined on an inner surface of the internal anvil portion cavity **135**. The external splines **123** and the internal splines **133** may engage with each other, locking the external anvil portion **120** and restricting its rotation with respect with the internal anvil portion **130**. The splines **123** and **133** allow for a transfer of the torque transmitted by the hammer **106** to the output shaft **125**. The internal splines **131** and the external splines **123** are configured to engage with each other. It should be understood that the number of splines may change in embodiments of the split anvil assembly **115**. The internal splines **131** and the external splines **123** may be shaped with square splines (tooth splines) or have differently shaped splines, including but not limited to radial slots, arc teeth, keyways, curvilinear splines, and/or triple square splines.

(25) Referring to FIG. 3, an anvil assembly **115** is shown having the lubrication port inlet **144** defined on a frontal end **129** of the output shaft **125**. The lubrication port inlet **144** is fitted with the grease port **145**. The lubrication passage **141** may extend longitudinally from the frontal end **129** to an anvil rear end **139**. In FIG. 3, the lubrication passage **141** splits the lubricant flow into the lubrication channels **142**. This front lubrication assembly **140** may be used in applications where the anvil assembly **115** is fixed within the hammercase **104**.

(26) FIGS. 4-6 show an impact tool having a ball-and-cam-type impact assembly **110**. The impact assembly **110** includes a cam shaft **150**, a bearing **151**, an impact bearing **154**, a hammer **106** and an anvil assembly **115**. The cam shaft **150** is driven for rotation about the longitudinal axis **100A** by the drive mechanism **105**. The cam shaft **150** includes a planetary gear carrier **153** for coupling to the drive mechanism **105**. The cam shaft **150** is coupled to the hammer **106** through the impact bearing **154**. The hammer **106** is rotatable over the bearing **151** and in turn drives rotation of the anvil assembly **115** about the longitudinal axis **100A**. The anvil assembly **115** includes the external anvil portion **120** and the internal anvil portion **130**.

(27) The cam shaft **150** and the hammer **106** each include a pair of opposed helical grooves **152** and **156**, respectively. The hammer grooves **156** have open ends facing the anvil assembly **115**. Thus, the cam shaft groove **152** is partially defined by a forward facing wall **152a** and a rearward facing wall **152b**, while the hammer groove **156** is partially defined by a forward facing wall **156a** and lacks a rearward facing wall. A pair of balls **154b** forming the impact bearing **154** couple the cam shaft **150** to the hammer **106**. Each ball **154b** is received in a race formed by the hammer groove **156** and the corresponding cam shaft groove **152**.

(28) A spring member **149** is disposed between the planetary gear carrier **153** and the hammer **106** to bias the hammer **106** away from the planetary gear carrier **153**. A forward-facing end of the hammer **106** includes a pair of hammer jaws **112** for driving rotation of the anvil assembly **115**. The anvil assembly **115** likewise includes a pair of anvil jaws **132** for cooperating with the hammer jaws **112**.

(29) The biasing force of the spring member **149** forces the hammer **106** away from the planetary gear carrier **153**. The forward-facing wall **156a** of the hammer groove **156** presses against a rearward portion of the balls **154**. This presses a forward portion of the balls **154b** against the rearward-facing surface **152b** of the cam shaft groove **152**. The balls **154b** are thereby trapped between the cam shaft **150** and the hammer **106** and couple the hammer **106** to the cam shaft **150**.

(30) In this embodiment, the front lubrication assembly **140** includes the lubrication port inlet **144** defined on the internal anvil cavity wall **130a**. The lubrication passage **141** extends to and through an anvil rear wall **130b**, where the anvil rear wall **130b** abuts with the cam shaft **150**. A cam shaft passage **161** is located within the cam shaft **150**. The cam shaft passage **161** is aligned with the lubrication passage **141** and may be parallel with the axis **100A**. The cam shaft passage **161** may include at least one cam shaft channel **162** having a cam shaft channel outlet **163**. The cam shaft channel **162** extends away from the cam shaft passage **161**. For example, the cam shaft channel **162**

may extend radially away from, or perpendicular to, the lubrication passage **141**. In other embodiments (not shown) the at least one cam shaft channel **162** may extend at an angle between zero degrees (0°) and ninety degrees (90°) with respect to the cam shaft passage **141** or the axis **100A**.

(31) The lubrication assembly **140** delivers the lubricant injected into the grease fitting **145** to the impact assembly **110**. In embodiments, the cam shaft channel outlet **163** is located proximate to or adjacent to the impact bearing **154**. The lubricant flow exits the cam shaft channel outlet **163** and lubricates the impact bearing **154** and the opposing helical grooves, the cam shaft groove **152** and the hammer groove **156**. Additionally, the axial repetitive motion of the hammer **106** with respect to the cam shaft **150** may also transport at least a portion of the lubricant flow to the at least one anvil jaw **132** and/or the at least one hammer jaw **112**. In other embodiments, both the lubrication passage **141** and the cam shaft passage **161** include a lubrication channel **142** and a cam shaft channel **162** extending radially away from their respective passages.

(32) In other embodiments, the split anvil assembly may define the front lubrication assembly **140** having the lubrication passage **141** extend along both the external anvil portion **120** and the internal anvil portion **130**. The lubrication port inlet **144** may be defined on a frontal end **129** of the output shaft **125** of the external anvil portion **120**. In this embodiment, a lubrication seal may be disposed between the external anvil portion **120** and the internal anvil portion **130**.

(33) The impact tool assembly **100** having a front lubrication port **145** may use interchangeable output shafts **125** having different drive diameters, extended anvils, or accessories such as socket extensions and socket adapters. For example, different embodiments of the anvil assembly **115** may have different sizes of output shaft **125**. The output shaft **125** of anvil assembly **115** may range from one-quarter of an inch ($\frac{1}{4}$ in.), to two and one-half inches ($2\frac{1}{2}$ in.). For example, the output shaft may be sized for drive sizes of $\frac{1}{4}$ in., $\frac{3}{8}$ in., $\frac{1}{2}$ in., $\frac{3}{4}$ in., 1 in. $1\frac{1}{2}$ in., and $2\frac{1}{2}$ in. It should be understood that these drive sizes are examples and not limiting to any sizes in metric and/or U.S. units.

(34) While the subject matter has been illustrated and described in detail in the drawings and foregoing description, the same is to be considered as illustrative and not restrictive in character, it being understood that only example embodiments have been shown and described and that all changes and modifications that come within the spirit of the subject matters are desired to be protected. In reading the claims, it is intended that when words such as “a,” “an,” “at least one,” or “one of a plurality of” are used there is no intention to limit the claim to only one item unless specifically stated to the contrary in the claim. Unless specified or limited otherwise, the terms “mounted” and “connected” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, and couplings. Further, “connected” is not restricted to physical or mechanical connections or couplings.

Claims

1. An impact tool comprising: a housing having a front end and a rear end, the housing configured to house a drive mechanism; an impact assembly configured to be driven by the drive mechanism about an axis extending from the front end to the rear end, the impact assembly including: a hammer having at least one hammer jaw; and a split anvil assembly including: an internal anvil portion fully disposed inside the housing, the internal anvil portion defining at least one anvil jaw and an internal anvil portion cavity, the at least one anvil jaw configured to periodically engage with the at least one hammer jaw to rotate the internal anvil portion about the axis; and an external anvil portion configured to be removably received within the internal anvil portion cavity and to engage with the internal anvil portion so that the external anvil portion rotates with the internal anvil portion, wherein the anvil assembly further includes a front lubrication assembly having a lubrication passage that extends longitudinally along the axis, the lubrication passage configured to

direct a lubricant from a lubrication port to the impact assembly.

2. The impact tool of claim 1, wherein the anvil assembly further includes at least one lubrication channel extending away from the lubrication passage, the at least one lubrication channel having a channel outlet located adjacent to at least one of the at least one anvil jaw or the at least one hammer jaw of the impact tool.

3. The impact tool of claim 2, wherein the at least one lubrication channel extends radially away from the lubrication passage.

4. The impact tool of claim 1, wherein the lubrication port is disposed within the internal anvil portion cavity.

5. The impact tool of claim 1, wherein the impact assembly further includes a cam shaft and an impact bearing, wherein the cam shaft is coupled to the hammer through the impact bearing, wherein the cam shaft includes at least one cam shaft channel extending away from a cam shaft passage, the at least one cam shaft channel having a cam shaft channel outlet adjacent to the impact bearing, and wherein the cam shaft channel outlet is configured to deliver lubricant to the impact bearing.

6. The impact tool of claim 5, wherein motion of the hammer with respect to the cam shaft directs lubricant to at least one of the at least one anvil jaw or the at least one hammer jaw of the impact tool.

7. An impact tool comprising: a housing having a front end and a rear end, the housing configured to house a drive mechanism; an impact assembly configured to be driven by the drive mechanism about an axis extending from the front end to the rear end, the impact assembly including: a hammer having at least one hammer jaw; and an anvil assembly having at least one anvil jaw and an output shaft, the at least one anvil jaw configured to periodically engage with the at least one hammer jaw to rotate the anvil assembly about the axis; wherein the anvil assembly further includes a front lubrication assembly having a lubrication passage that extends longitudinally along the axis, the lubrication passage configured to direct a lubricant from a lubrication port to the impact assembly; and wherein the anvil assembly comprises: an internal anvil portion fully disposed inside the housing, the internal anvil portion defining an internal anvil portion cavity; and an external anvil portion configured to be removably received within the internal anvil portion cavity, the external anvil portion configured to be partially received within the housing and to engage with the internal anvil portion so that the external anvil portion rotates with the internal anvil portion.

8. The impact tool of claim 7, wherein the anvil assembly further includes at least one lubrication channel extending away from the lubrication passage, the at least one lubrication channel having a channel outlet located adjacent to at least one of the at least one anvil jaw or the at least one hammer jaw of the impact tool.

9. The impact tool of claim 8 wherein the at least one lubrication channel extends perpendicularly away from the lubrication passage.

10. The impact tool of claim 8, wherein the lubrication port is disposed within the internal anvil portion cavity.

11. The impact tool of claim 7, wherein the impact assembly further includes a cam shaft and an impact bearing, wherein the cam shaft is coupled to the hammer through the impact bearing, wherein the cam shaft includes at least one cam shaft channel extending away from a cam shaft passage, the at least one cam shaft channel having a cam shaft channel outlet adjacent to the impact bearing, and wherein the cam shaft channel outlet is configured to deliver lubricant to the impact bearing.

12. The impact tool of claim 11, wherein motion of the hammer with respect to the cam shaft directs lubricant to at least one of the at least one anvil jaw or the at least one hammer jaw of the impact tool.

13. An impact tool comprising: a housing having a front end and a rear end, the housing configured

to house a drive mechanism; an impact assembly configured to be driven by the drive mechanism about an axis extending from the front end to the rear end, the impact assembly including: a hammer having at least one hammer jaw; and a split anvil assembly including: an internal anvil portion disposed inside the housing, the internal anvil portion defining at least one anvil jaw and an internal anvil portion cavity, the at least one anvil jaw configured to periodically engage with the at least one hammer jaw to rotate the internal anvil portion about the axis; and an external anvil portion configured to be removably received within the internal anvil portion cavity and to engage with the internal anvil portion so that the external anvil portion rotates with the internal anvil portion, wherein the anvil assembly further includes a front lubrication assembly having a lubrication passage disposed within the internal anvil portion cavity, the front lubrication assembly extending longitudinally along the axis, the lubrication passage configured to direct a lubricant from a lubrication port to the impact assembly, the lubrication port being disposed within the internal anvil portion cavity.

14. The impact tool of claim 13, wherein the anvil assembly further includes at least one lubrication channel extending away from the lubrication passage, the at least one lubrication channel having a channel outlet located adjacent to at least one of the at least one anvil jaw or the at least one hammer jaw of the impact tool.

15. The impact tool of claim 14, wherein the at least one lubrication channel extends perpendicularly away from the lubrication passage.

16. The impact tool of claim 13, wherein the impact assembly further includes a cam shaft and an impact bearing, wherein the cam shaft is coupled to the hammer through the impact bearing, wherein the cam shaft includes at least one cam shaft channel extending away from a cam shaft passage, the at least one cam shaft channel having a cam shaft channel outlet adjacent to the impact bearing, and wherein the cam shaft channel outlet is configured to deliver lubricant to the impact bearing.

17. The impact tool of claim 16, wherein motion of the hammer with respect to the cam shaft directs lubricant to at least one of the at least one anvil jaw or the at least one hammer jaw of the impact tool.
